Acceleration means velocity is changing

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Acceleration / ES: aceleración / PT: aceleração

The rate velocity changes: how much velocity is gained or lost each second.

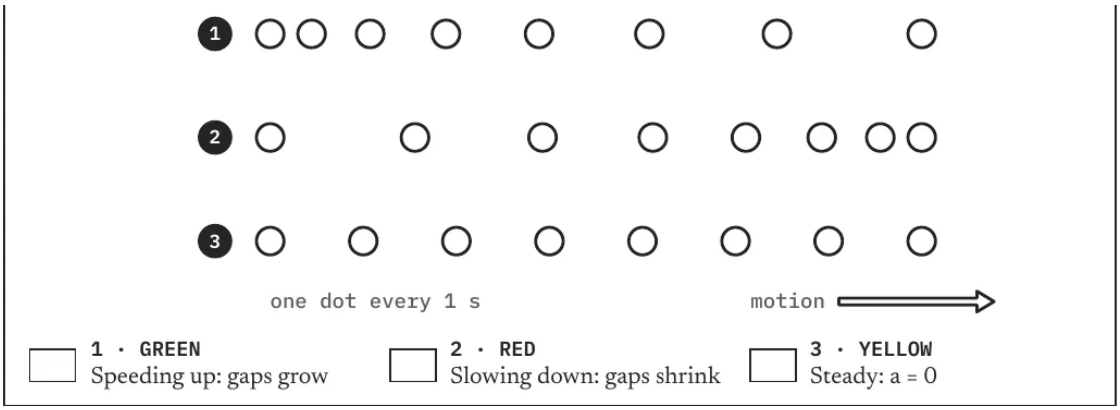
Also on your note page: Speeding up, Slowing down, Turning.

B READ THE RELATIONSHIP

Acceleration tells how quickly velocity changes.

Average acceleration = change in velocity ÷ time

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart's velocity changes from 0 to 8 m/s in 4 s. Find its acceleration.

Given $v_i = 0$ m/s $v_f = 8$ m/s $\Delta t = 4$ s

Rule $a = \Delta v \div \Delta t$

Work $\Delta v = 8 - 0 = 8$ m/s $a = 8 \div 4$

Answer 2 m/s² gains 2 m/s each second

E SAVE FOR THE FINAL MEETING

If XJ's velocity readings change from one night to the next, it is accelerating, and a forecast that assumes steady velocity will miss.

Read the physics: Acceleration

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Compare velocity before and after. A change in speed or a turn is a change in velocity. Divide the signed change in velocity by the time it took.

02 HOW IT WORKS

Read m/s² as a change in meters per second each second. Look at dot spacing at equal times: growing gaps mean faster motion; shrinking gaps mean slower motion.

03 WHAT TO CHECK

Compare velocity and acceleration. In a straight line, matching directions mean speed grows. Opposite directions mean speed falls until the object stops or reverses.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart's velocity changes from 0 to 8 m/s in 4 s. Find its acceleration. 2 m/s² gains 2 m/s each second.

05 CONNECT TO XJ

If XJ's velocity readings change from one night to the next, it is accelerating, and a forecast that assumes steady velocity will miss.

SAY IT BEFORE YOU WRITE

Can acceleration be negative while speed increases? The velocity changes from ___ to ___. Each second it changes by ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Acceleration / ES: aceleración / PT: aceleração

Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, acceleration means …	A cart goes around a curve at steady speed. What changes? The speed stays the same, but the ___ changes.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Draw the before and after velocity arrows (m/s). Mark the change in velocity (m/s). Divide by time to label the acceleration (m/s²).

Labels you can use: before · after · change.

Before	After

Tell a partner: The velocity changes from ___ to ___. Each second it changes by ___.

On assignment: Acceleration

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two signed XJ velocities are −2 and −6 km/s, two seconds apart. The control room must choose its speed alert.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Telemetry engineer**

Send the average acceleration with its sign and unit. Sketch before and after velocity arrows so the shift lead can confirm the alert.

☐ **Replay commentator**

Narrate what changed over two seconds. Have your producer show which arrow proves whether XJ sped up or slowed down.

Word bank: before · after · change.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”